



Mobile CPR Point Detector

Aplikasi React Native Expo untuk mendeteksi titik kompresi CPR menggunakan kamera smartphone secara real-time dengan AI pose detection (MoveNet).



Features

✓ Real-time CPR Detection

- Deteksi pose menggunakan MoveNet model
- Kalkulasi otomatis titik kompresi CPR
- Visualisasi marker di atas live camera

✓ Simple & Clean UI

- Minimalist interface
- FPS counter
- Confidence indicator
- Skeleton overlay (toggle on/off)

✓ Mobile Optimized

- Front camera support
- Smooth performance
- Low latency detection



Prerequisites

- Node.js (v16 atau lebih baru)
- npm atau yarn
- Expo CLI: `npm install -g expo-cli`
- Smartphone Android/iOS dengan kamera
- Expo Go app di smartphone ([Android](#) | [iOS](#))

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Installation

1. Install Dependencies

```
cd mobile-cpr-detector
npm install
```

2. Start Development Server

```
npm start
```

atau

```
expo start
```

3. Run on Device

Setelah development server running:

Opsi A: Scan QR Code (Recommended)

1. Buka Expo Go app di smartphone
2. Scan QR code yang muncul di terminal/browser
3. App akan loading dan running di device

Opsi B: Run via Emulator

```
# Android
npm run android

# iOS (Mac only)
npm run ios
```



Usage Instructions

1. Grant Camera Permission

- Saat pertama kali buka app, izinkan akses kamera

2. Posisikan Tubuh

- Hadap kamera (frontal view)
- Tampilkan upper body (kepala sampai pinggang)
- Pastikan pencahayaan cukup

3. Lihat CPR Point

- Yellow/Green marker = CPR compression point
- Muncul di tengah dada ketika pose terdeteksi
- Confidence level ditampilkan di bottom bar

4. Toggle Skeleton (Optional)

- Tap tombol "Show Skeleton" untuk melihat full pose keypoints
- Tap lagi untuk hide skeleton



Troubleshooting

Camera Permission Denied

Settings > Apps > Expo Go > Permissions > Camera > Allow

Model Loading Error

- Pastikan koneksi internet stabil (first time load)
- Clear app cache: hapus dan install ulang Expo Go
- Check logs: `expo start --dev-client`

Slow Performance

- Close other apps
- Reduce camera resolution (edit `cameraTextureHeight/Width` di `CameraScreen.js`)
- Test di device yang lebih powerful

CPR Point Not Detected

- Pastikan upper body terlihat jelas
- Improve lighting
- Stand further dari camera (full upper body visible)
- Check confidence threshold (default: 0.3)



Project Structure

```
mobile-cpr-detector/
├── App.js                # Main app entry
├── components/
│   ├── CameraScreen.js  # Main camera & detection logic
│   └── CPROverlay.js     # SVG overlay untuk CPR marker
├── utils/
│   └── cprCalculation.js # CPR point calculation utility
├── package.json          # Dependencies
├── app.json              # Expo configuration
└── README.md             # This file
```

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CPR Calculation Formula

Titik CPR dihitung menggunakan formula geometris:

$$\text{CPR_point} = \text{shoulder_midpoint} + \alpha \times (\text{hip_midpoint} - \text{shoulder_midpoint})$$

Dimana:

- $\alpha = 0.42$ (42% distance dari shoulders ke hips)
- `shoulder_midpoint` = midpoint antara left & right shoulder

- `hip_midpoint` = midpoint antara left & right hip

Ini menghasilkan titik di area sternum (center chest, lower sternum) yang merupakan lokasi ideal untuk kompresi CPR.

Customization

Mengubah Marker Color

Edit `components/CPROverlay.js`:

```
const { level, color } = getConfidenceLevel(cprPoint.confidence);  
// Modify color logic di utils/cprCalculation.js
```

Adjust Confidence Threshold

Edit `utils/cprCalculation.js`:

```
const CONFIDENCE_THRESHOLD = 0.3; // Change to 0.4, 0.5, etc.
```

Camera Resolution

Edit `components/CameraScreen.js`:

```
<TensorCamera  
  cameraTextureHeight={1920} // Reduce for better performance  
  cameraTextureWidth={1080}  
  resizeHeight={192} // Model input size  
  resizeWidth={192}  
</>
```

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Performance Tips

1. Optimize for Production

```
expo build:android  
expo build:ios
```

2. Enable Hermes (Android)

- Sudah enabled by default di Expo SDK 51

3. Reduce Bundle Size

- Only import needed TensorFlow.js modules
- Remove unused dependencies

Privacy & Security

⚠ Important Notes:

- Semua processing dilakukan **locally on-device**
- Tidak ada data dikirim ke server
- Tidak ada recording/storing video
- Camera access hanya saat app aktif

License

MIT License - Free to use and modify

Credits

- **MoveNet Model:** Google TensorFlow Team
- **TensorFlow.js:** TensorFlow Team
- **Expo:** Expo Team

🔧 Support

Jika ada issue atau pertanyaan:

1. Check troubleshooting section
2. Review Expo documentation: <https://docs.expo.dev>
3. TensorFlow.js docs: <https://www.tensorflow.org/js>

Made with ❤ for CPR Training & Education

Quick Start Summary

```
# 1. Install dependencies
cd mobile-cpr-detector
npm install

# 2. Start server
npm start

# 3. Scan QR code dengan Expo Go app
# 4. Grant camera permission
# 5. Position upper body facing camera
# 6. See CPR point detected! 🎯
```

That's it! Simple and ready to use! 📱 ✨